

Project 3: Online E-commerce Website using Agile

1. Abstract

The Online E-commerce Website project aims to build a scalable and interactive platform for online retail. Developed using **Agile methodology**, the project focuses on delivering high-value features such as product search, shopping cart management, and secure checkout through iterative Sprints. This approach ensures that the platform can adapt to changing market trends and user feedback during development. The system provides a seamless shopping experience for customers while offering an administrative backend for inventory and order management.

2. Introduction

- **2.1 Introduction:** This project involves the development of a comprehensive web-based storefront for global commercial transactions.
 - **2.2 Problem Identification:** Traditional retail and poorly designed websites often suffer from slow loading times, lack of mobile responsiveness, and complex checkout processes that lead to cart abandonment.
 - **2.3 Need of the project:** There is a growing need for a secure, fast, and highly available e-commerce solution that supports personalized user experiences and multiple payment methods.
 - **2.4 Project Scheduling:** The development is structured into Sprints: Sprint 1 (User Authentication & Database), Sprint 2 (Product Catalog & Search), Sprint 3 (Cart & Payment Integration), and Sprint 4 (Order Tracking & Admin Panel).
 - **2.5 Objectives:** To provide an intuitive shopping interface, ensure data security for transactions, and implement an automated inventory update system.
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3. Software Requirement Specification

- **3.1 Purpose:** To provide a robust digital marketplace for buyers and sellers.

- **3.2 Scope:** Includes user registration, product filtering, shopping cart, payment gateway integration, and order history.
 - **3.3 Hardware Requirement / Software Requirement (minimum):**
 - **Hardware:** Intel i5 Processor, 8GB RAM, 50GB available disk space.
 - **Software:** Java JDK 17, MySQL Database, Apache Tomcat Server, Windows/Linux OS.
 - **3.4 Tools:** Eclipse/IntelliJ IDE, Selenium WebDriver for automation, Maven for dependency management, and Git for version control.
 - **3.5 Software Process Model: Agile Scrum Model,** facilitating rapid iterations and continuous deployment.
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4. System Design

- **4.1 Data Dictionary:** Defines elements like Product_ID, Category_Name, Stock_Quantity, and Order_Total.
 - **4.2 ER Diagram:** Illustrates the relationships between Customer, Product, Order, Category, and Payment.
 - **4.3 DFD (Data Flow Diagram):** Shows the movement of data from product selection to payment authorization and shipping status updates.
 - **4.4 Class Diagram:** Represents the blueprint of the system, including classes like ProductService, CartManager, and PaymentProcessor.
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5. Implementation

- **5.1 Program Code:** Built using Java technologies for the backend and Selenium scripts to automate the testing of the "Add to Cart" and "Checkout" functionalities.
 - **5.2 Output Screens:** Snapshots of the Home Page, Product Search Results, and the Order Confirmation Screen.
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6. Testing

- **6.1 Test Data:** Sets of valid and invalid product IDs, various credit card test numbers, and empty cart scenarios.
 - **6.2 Test Result:** Documentation confirming successful execution of regression tests and performance tests under concurrent user access.
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7. User Manual

- **7.1 How to use project guidelines:** Instructions for users on how to register, apply discount coupons, and track their orders.
 - **7.2 Screen Layouts and Description:** Visual guide explaining the function of the "Wishlist" icon, "Filter" sidebar, and "Compare" tool.
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8. Project Applications and Limitations

- **Applications:** Suitable for B2C retail businesses, fashion brands, and electronic stores.
 - **Limitations:** Security is dependent on the third-party payment provider; requires consistent maintenance for catalog updates.
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9. Conclusion and Future Enhancement

- **Conclusion:** The Agile-based development ensured the delivery of a user-centric platform that meets modern retail standards.
 - **Future Enhancement:** Integration of Augmented Reality (AR) for "Virtual Try-on" features and AI-powered recommendation engines.
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10. Bibliography & References

- References include the Agile Manifesto, Selenium documentation, and Java Web Development guides.